NIST UNIVERSITY

Department of Computer Science and Engineering

DESIGN AND ANALYSIS OF ALGORITHMS (DAA) · B.Tech 3rd Semester Project

EV CHARGING STATION PLACEMENT AND ROUTE PLANNING SYSTEM

Course:	Design and Analysis of Algorithms
Programme:	B.Tech in Computer Science & Engineering (Data Science)
Academic Year:	2026–27
Place:	NIST University, Berhampur, Odisha

Submitted To

Faculty Guide: Prof. _____
Department of Computer Science and Engineering
NIST University

Submitted By

Om Prakash Sahu
Roll No: 202556339

B.Tech CSE (Data Science), 3rd Semester
Prasant Nayak
Roll No: 202556393
B.Tech CSE (Data Science), 3rd Semester

Certificate

This is to certify that the project report entitled **EV Charging Station Placement and Route Planning System** has been prepared by **Om Prakash Sahu (202556339)** and **Prasant Nayak (202556393)**, students of B.Tech CSE (Data Science), 3rd Semester, NIST University, Berhampur, Odisha, for the academic session 2026–27.
The project demonstrates the implementation of Kruskal's MST, 0/1 Knapsack DP, and Dijkstra's Shortest Path algorithms in a professional web application environment.

Internal / External Viva-Voce Examiner Evaluation
Evaluated on Date: _____
Remarks:

_____

_____
Signature of Examiner & Date

_____

Declaration

We hereby declare that this project report on **EV Charging Station Placement and Route Planning System** is the result of our original academic work for the Design and Analysis of Algorithms course. The algorithms (Kruskal, Knapsack, Dijkstra) were implemented manually by the students. No fabricated results are included; experimental results are generated from the implemented system.

Student 1	Student 2	Guide / Faculty
Om Prakash Sahu Roll No: 202556339 B.Tech CSE (Data Science), 3rd Sem Signature: _____	Prasant Nayak Roll No: 202556393 B.Tech CSE (Data Science), 3rd Sem Signature: _____	Name: _____ Date: _____ Signature: _____

Statement of Originality
The work presented herein is original and adheres to NIST University academic standards.

Acknowledgement

We express our sincere gratitude to the Department of Computer Science and Engineering, NIST University, for providing the laboratory resources and academic environment. We thank the open-source community for technologies like Python, Flask, and Leaflet. Special thanks to our classmates for technical support during algorithm testing.

Institutional & Academic Submission Summary

Project Attribute	Institutional & Academic Details
Project Title	EV Charging Station Placement and Route Planning System
Subject	Design and Analysis of Algorithms
Programme	B.Tech CSE (Data Science)
Semester	3rd
Institution	NIST University
Academic Session	2026–27
Students	Om Prakash Sahu & Prasant Nayak
Roll Numbers	202556339 & 202556393
Core Algorithms	Kruskal, Dijkstra, 0/1 Knapsack

Abstract

This report documents the design and implementation of an EV Charging Station Placement and Route Planning System. The system models road networks as weighted graphs and utilizes Kruskal's Minimum Spanning Tree for network analysis, 0/1 Knapsack Dynamic Programming for budget-constrained station selection, and Dijkstra's Shortest Path for route calculation. The application integrates real-world data sources and provides a professional product-grade UI/UX interface.

Executive Technical Profile

Technical Dimension	Implementation Specification
Application Type	Deployable Web Application
Backend	Python Flask
Frontend	HTML/CSS/JavaScript (Professional UI)
Graph Model	Weighted Graph
MST Algorithm	Kruskal (Manual)
Station Optimization	0/1 Knapsack (Manual DP)
Route Algorithm	Dijkstra (Manual)
Visualization	Interactive Map (Leaflet/MapLibre)
Deployment	Render (Gunicorn)
Testing	Automated Unit + Integration Tests

Keywords
EV infrastructure, Graph Algorithms, Kruskal, Dijkstra, 0/1 Knapsack, Dynamic Programming, Minimum Spanning Tree, Shortest Path, Route Planning, Optimization.

Table of Contents

PART 1 — FRONT MATTER
Cover Page 1
Certificate 2
Declaration 3
Acknowledgement 4
Abstract 5
Table of Contents 6
PART 2 — PROJECT INTRODUCTION AND REQUIREMENT ANALYSIS
1. Introduction 7
2. Problem Statement and Objectives 8
3. Scope and Feasibility Analysis 9
4. Software and Hardware Requirements 10
PART 3 — ALGORITHMIC DESIGN AND SYSTEM MODELING
5. System Architecture and Data Flow 11
6. Use-Case Diagram and Functional Specifications 12
7. Graph Model and Network Representation 13
8. Algorithm Design 14
9. Algorithm Complexity Analysis 15
PART 4 — IMPLEMENTATION
10. Kruskal MST Implementation 16
11. 0/1 Knapsack Implementation 17
12. Dijkstra Shortest Path Implementation 18
13. Integrated Optimization Pipeline 19
PART 5 — APPLICATION AND RESULTS
14. Web Application Interface 20
15. Graph Visualization and Network Analysis 21
16. Charging Station Optimization Results 22
17. Route Planning Results 23
PART 6 — TESTING, CONCLUSION AND REFERENCES
18. System Testing 24

19. Results and Discussion 25
20. Limitations and Future Scope 26
21. Conclusion 27
22. References 28
List of Figures 29
List of Tables 30

1. Introduction

The adoption of Electric Vehicles (EVs) necessitates robust charging infrastructure. This project models road networks as weighted graphs to optimize station placement under budget constraints and calculate efficient routes using classical DAA algorithms. The system serves as a decision-support platform for urban planners and infrastructure analysts.

8. Algorithm Design

8.1 Kruskal

Objective: Find Minimum Spanning Tree (MST) of the road network. The manual implementation uses Union-Find with path compression and union by rank.

```
def kruskal_mst(nodes, edges): ...
```

8.2 0/1 Knapsack

Objective: Maximize benefit within budget constraint. Dynamic Programming table construction and backtracking are used.

```
def knapsack(stations, budget): ...
```

8.3 Dijkstra

Objective: Calculate shortest path between source and destination. Priority queue and relaxation are used.

```
def dijkstra(graph, source, destination): ...
```

9. Complexity Analysis

Algorithm	Time Complexity	Space Complexity	Main Operation
Kruskal	O(E log E)	O(V+E)	Edge sorting + Union-Find
Dijkstra	O((V+E) log V)	O(V+E)	Relaxation + Min Heap
0/1 Knapsack	O(nW)	O(nW)	DP table

22. References

- Cormen et al., Introduction to Algorithms
- Kleinberg & Tardos, Algorithm Design
- Flask Official Documentation (https://flask.palletsprojects.com)
- Python Official Documentation (https://python.org)
- Render Deployment Documentation (https://render.com)